
Software Requirements Specification

for

FaceRoll - Face Recognition Attendance System

Version 1.3 approved

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1. Introduction

1.1 Purpose

The Purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements for FaceRoll - Face Recognition Attendance System, Version 1.0. This document will provide the specified scope, behavior, constraints, and requirements of the FaceRoll system, which is designed to automate classroom attendance in higher education environments using facial recognition technology.

The SRS covers the core functionality of the FaceRoll system, including student attendance check-in, instructor attendance management, data storage for face recognition, and optional integration with the Canvas Learning Management System. The requirements described in this document will serve as a baseline for system design, development, testing, and evaluation throughout the project lifecycle.

1.2 Intended Audience and Reading Suggestions

This document is intended for the following audiences:

- Project Developers: To understand system requirements, constraints, and expected behavior during implementation.
- Project Managers and Team Leads: To follow the scope, dependencies, and progress throughout the development process.
- Instructors and project sponsors: To review and validate that the system requirements align with the institutional needs.
- Testers and Reviewers: To verify that the implemented features meet the specified requirements.

This document is organized into sections that progressively describe the system. Readers unfamiliar with the project should begin with sections 1 and 2 to gain an understanding of the system and its context. Developers and testers should focus on sections 3 and 4 for detailed system features and requirements. Non-functional requirements and constraints are described in later sections for reference during implementation and evaluation.

1.3 Product Scope

FaceRoll is a face recognition-based attendance system developed to reduce the time and effort required for manual attendance tracking in classroom settings. The system uses a camera-enabled environment to perform real-time face detection and recognition, allowing students to be identified and marked present without requiring manual input during the attendance process.

FaceRoll provides instructors with a browser-based dashboard that allows them to view attendance records by session, search and filter student data, and export attendance reports for administrative use. The system includes a manual override capability that allows instructors to modify attendance records when necessary.

Students interact with the system through a mobile application to log in, enroll their faces, perform attendance check-ins during active sessions, and view attendance history.

FaceRoll securely stores attendance data and facial recognition embeddings, supporting a privacy-conscious design that avoids storing raw images of students. The system operates independently of campus infrastructure and is not required to work on campus Wi-Fi networks. Additionally, FaceRoll supports integration with Canvas LMS through its API, allowing attendance data to be transferred when configured and authorized, without modifying existing institutional systems.

This system is intended to improve attendance accuracy, reduce administrative workload for instructors, and provide a flexible, secure, and efficient attendance solution.

In Scope: The system includes automated face recognition during attendance sessions, an instructor dashboard for viewing and managing attendance record, manual override functionality for correcting attendance data, secure storage of attendance information and facial embeddings, a mobile application for students to enroll their face, attendance check-ins, view attendance history and verify recognition status, locally hosted deployment with no external biometric data within cloud services, and integration with Canvas LMS through its API.

Out of Scope: The system does not include operation dependent on campus Wi-Fi networks, integration with learning management systems other than Canvas, real-time surveillance or tracking outside of the classroom, storage or sharing raw biometric data beyond what is necessary for recognition, provision of physical hardware such as cameras or classroom devices, or advanced analytics such as predictive attendance modeling.

References

- [1] Project Charter - FaceRoll: Face Recognition Attendance System, Team Cowboys, Version 1.0, January 2026.
- [2] Canvas Learning Management System API Documentation, Instructure, Available online
- [3] Google. Firebase Documentation. Available online.
- [4] U.S. Department of Education, Family Educational Rights and Privacy Act (FERPA). Available online.

2. Overall Description

2.1 Product Perspective

FaceRoll is a brand-new, stand-alone attendance management system developed for higher education classrooms. It serves as a supporting system that automates attendance but does not take the place of Canvas LMS.

The system consists of the following major components:

- Cloud Firestore
- Student Mobile app (React Native mobile client)

- Instructor Dashboard (Web Interface)
- Python-based facial recognition backend
- Optional Canvas API Integration

2.2 Product Functions

At a high level, FaceRoll will:

- Capture facial data during student enrollment
- Detect and recognize student faces in real time
- Automatically mark students as Present or Late
- Allow instructors to manually override attendance
- Store attendance records securely
- Generate attendance reports

2.3 User Classes and Characteristics

Instructors (Primary User Class)

- Frequency of use: Daily
- Technical skill level: Moderate
- Permissions:
 - Start attendance session
 - View real-time attendance
 - Override attendance

- Export reports
- Priority: Highest priority user group

Students

- Frequency of use: Daily
- Technical skill level: Low to moderate
- Permissions:
 - Participate in a facial recognition check-in
- Limited direct system interaction

System Administrators

- Frequency of use: Occasional
- Technical skill level: High
- Permissions:
 - Manage system configuration
 - Manage user enrollment
 - Maintain database

2.4 Operating Environment

FaceRoll operates as a hybrid system consisting of two primary environments

1. An On-Premises Edge Processing Environment responsible for all biometric processing
2. A Cloud-Based Application Environment responsible for attendance management, dashboard access, and Canvas integration.

Attendance Capture Device Requirements:

- **Operating System:** Windows
- **AI Runtime:** Python 3.10 or 3.11
- **Face Detection Library:** OpenCV 4.x
- **Face Recognition Model:** DeepFace with a FaceNet-based model
- **Hardware:**
 - USB or integrated HD camera (minimum 720p)
 - Minimum 8GB RAM recommended
 - CPU capable of real-time processing

Responsibilities of the Attendance Capture Device:

- Capture live camera frames
- Perform face detection
- Generate facial embeddings
- Compare embeddings against locally stored enrollment data
- Determine Attendance status (Present, Late, Absent)
- Store facial embeddings locally in an encrypted format
- Transmit only attendance metadata to the cloud environment

Mobile Platform

- **Android:** Android 12 (API level 31) or higher
- **iOS:** iOS 16 or higher
- **Mobile Framework:** React Native (Expo SDK 50+)
- **Programming Language:** TypeScript

The application requires:

- Front-facing camera access
- Internet connectivity (Wi-Fi, LTE, or 5G)
- Minimum 4 GB RAM device recommended

Backend Environment

The cloud environment shall be used exclusively for non-biometric data management and system coordination.

- **Cloud Platform:** Google Firebase

- **Recognition Backend Runtime:** Python 3.10 or 3.11
- **Cloud Functions:** Firebase Cloud Functions
- **Database:** Cloud Firestore
- **Authentication:** Firebase Authentication
- **Recognition Backend Framework:** Flask
- **Recognition Backend Libraries:** DeepFace, OpenCV

External System Environment

FaceRoll integrates with the Canvas Learning Management System Sandbox environment.

- **Communication Protocol:** HTTPS
- **API Standard:** Canvas REST API
- **Authentication:** OAuth 2.0 / API Token

FaceRoll must coexist with institutional Canvas configurations without modifying Canvas core services.

2.5 Design and Implementation Constraints

- **Cloud Infrastructure Constraint**
The system must use Google Firebase as the hosting and database platform.
- **Mobile Application Constraint**
The system must be implemented as a cross-platform mobile application. No desktop or standalone web-only version will be delivered in Version 1.0.
- **Canvas Integration Constraint**
All LMS communication must occur through the official Canvas REST API. Direct database access to Canvas is prohibited.
- **Security and Regulatory Constraint**
The system must comply with FERPA regulations regarding student data privacy.

Biometric data must be encrypted at rest and in transit.

- **Biometric Storage Constraint**
Facial embeddings must be stored in an encrypted format.
Raw facial images shall not be permanently stored.
- **Communication Protocol Constraint**
All communications must use HTTPS with TLS 1.2 or higher encryption.
- **Technology Stack Constraint**
Backend logic must be implemented using Firebase
AI processing must be implemented using Python.
Database services must use Cloud Firestore.

2.6 User Documentation

- **User Manual (PDF Format)**
Comprehensive guide describing system features, user roles, facial enrollment process, attendance session management, and Canvas synchronization procedures.
- **Instructor Guide (PDF Format)**
Focused documentation for instructors covering session setup, attendance override, and Canvas export.
- **Student Quick Start Guide (PDF Format)**
Step-by-step instructions for login and check-in procedures.
- **In-App Help System**
Context-sensitive guidance embedded within the mobile application.

All documentation will:

- Be delivered electronically
- Use clear, non-technical language
- Follow standard software documentation structure
- Be accessible to all user roles

2.7 Assumptions and Dependencies

Assumptions

- A1. Users will have consistent internet connectivity during attendance sessions.
 - A2. Mobile devices used for check-in will support camera access and modern operating systems.
 - A3. Institutional approval has been granted for Canvas Sandbox integration.
 - A4. Students will provide consent for biometric enrollment where required.
 - A5. The AI recognition model will achieve acceptable accuracy within defined performance thresholds.
-

Dependencies

- D1. The system depends on the continued availability of Google Firebase services.
- D2. The system depends on the continued availability of the Canvas Sandbox API.
- D3. The system depends on the proper configuration of OAuth credentials for Canvas access.
- D4. The AI recognition service must remain operational and accessible during attendance sessions.
- D5. Mobile operating system updates must not introduce breaking changes to camera access APIs.

3. External Interface Requirements

3.1 User Interfaces

1. Instructor Dashboard

- Web-based interface
- Displays:
 - Course list
 - Attendance session controls
 - Real-time attendance list

- Override buttons (Present/Late/Absent)
- Export attendance (CSV)

Standard UI Elements:

- Top navigation bar
- Logout button
- Help button
- Error messages are displayed in red banners
- Confirmation dialogs for overrides

2. Student Check-In Interface

- Simple camera interface
- Displays:
 - Live webcam feed
 - Recognition status (Recognized / Not Recognized)
 - Success confirmation message

Design Guidelines:

- Minimal interface
- Large readable text
- Mobile-friendly responsive layout

3.2 Hardware Interfaces

The system interfaces with:

Mobile Device Camera

- Uses native camera APIs
- Must support:
 - Minimum 720p resolution
 - Real-time frame capture

Mobile Device Hardware

- Front-facing camera
- Internet connectivity (Wi-Fi or LTE/5G)

3.3 Software Interfaces

FaceRoll interfaces with internal backend services, database systems, authentication services, and the Canvas Sandbox environment to support attendance synchronization and course management.

Mobile Operating System Interface

The FaceRoll mobile application operates on:

- Android 12 or higher
- iOS 16 or higher

The application interfaces with the mobile operating system to access:

- Device camera for facial recognition
- Secure token storage
- Network communication services

Authentication Service Interface

FaceRoll interfaces with Firebase Authentication for user identity verification and role management.

The authentication service provides:

- Secure login validation
- Session token generation
- Role-based access control

Database Interface

FaceRoll interfaces with Cloud Firestore for persistent data storage.

The database stores:

- User profiles
- Course mappings
- Enrollment records
- Attendance session data
- Attendance logs

Backend API Interface

FaceRoll interfaces with a Firebase backend service responsible for system logic and validation.

The backend service performs:

- Attendance session management
- Check-in validation
- Duplicate attendance prevention
- Role authorization enforcement

- Communication coordination between AI recognition and Canvas integration

All communication uses:

- HTTPS protocol
- RESTful API architecture
- JSON-formatted payloads

Face Recognition Service Interface

FaceRoll interfaces with a Python-based facial recognition service.

The recognition service performs:

- Face detection
- Feature embedding extraction
- Matching against stored embeddings

Data transmitted to the service includes:

- Captured facial image data
- Session identifier

Data returned includes:

- Matched student identifier
- Confidence score
- Recognition result status

Canvas Sandbox Interface

FaceRoll interfaces directly with the Canvas Sandbox environment using the official Canvas REST API.

This interface enables:

- Retrieval of course data from Canvas
- Retrieval of enrolled student rosters
- Optional synchronization of validated attendance metadata with Canvas when authorized by the instructor.

Communication characteristics:

- HTTPS protocol
- OAuth 2.0 or API token authentication
- JSON data exchange format

Attendance records transmitted to Canvas include:

- Student Canvas ID
- Course ID
- Attendance status (Present, Late, Absent)
- Timestamp

FaceRoll does not directly access Canvas databases. All interactions occur through the official Canvas API endpoints provided within the Sandbox environment.

3.4 Communications Interfaces

FaceRoll operates as a mobile application that communicates with backend services, AI recognition services, database systems, and the Canvas Sandbox environment using standard internet communication protocols.

All network communication within the system shall use the HTTPS protocol. Secure Sockets Layer (SSL/TLS 1.2 or higher) shall be enforced to ensure encrypted data transmission between all system components.

Client–Server Communication

Communication between the mobile application and backend server shall:

- Use HTTPS over TCP/IP
- Follow RESTful API architecture
- Transmit data in JSON format
- Require authenticated session tokens

All user requests, including login, attendance check-in, session management, and attendance retrieval, shall be transmitted as secure HTTP requests.

No unencrypted HTTP communication shall be permitted.

Backend–AI Service Communication

Communication between the backend service and the facial recognition service shall:

- Use HTTPS
- Transmit image data and session identifiers securely
- Exchange responses in JSON format

The AI service shall return recognition results, including matched user identifiers and confidence scores. Biometric data shall not be transmitted without encryption.

Backend–Database Communication

Communication between backend services and Firebase Firestore shall:

- Use secure HTTPS connections
- Use authenticated service credentials
- Transmit structured JSON objects

Direct client-to-database communication shall not be permitted.

Backend–Canvas Sandbox Communication

Communication between FaceRoll and the Canvas Learning Management System Sandbox environment shall:

- Use HTTPS
- Use OAuth 2.0 or API token authentication
- Follow Canvas REST API specifications
- Exchange data in JSON format

Attendance data transmitted to Canvas shall include:

- Student Canvas identifier
- Course identifier
- Attendance status
- Timestamp

Biometric data shall never be transmitted to Canvas.

Message Formatting Standards

All inter-service communication shall:

- Use UTF-8 encoding
- Use JSON for structured data exchange
- Include standardized HTTP response codes
- Include structured error messages when failures occur

Data Synchronization Mechanisms

The system shall operate using a request-response communication model.

Attendance synchronization with Canvas shall occur:

- After backend validation
- Upon instructor confirmation
- Through secure API calls

Security and Encryption Requirements

4. All communications shall use TLS 1.2 or higher encryption.
5. Authentication tokens shall be securely transmitted and validated.
6. Biometric image data shall be encrypted during transmission.
7. The system shall comply with FERPA data protection standards.

8. System Features

8.1 User Authentication and Role Management

8.1.1 Description and Priority

This feature enables secure authentication of instructors, administrators, and students using Firebase Authentication. It enforces role-based access control to restrict system functionality based on user type.

8.1.2 Stimulus/Response Sequences

1. The instructor opens the login screen.
2. Instructor enters email and password.
3. The system validates credentials using Firebase authentication.
4. If valid, the instructor dashboard loads.
5. If invalid, the system displays an error message and prevents access.

For student log-in/check-in:

1. Student opens a mobile app
2. Student authenticates using their account
3. System verifies the session
4. Student gains access to the camera interface

8.1.3 Functional Requirements

REQ-1: The system shall require authenticated login before granting dashboard access.

REQ-2: The system shall assign a role (Instructor, Student, Administrator) upon successful login.

REQ-3: The system shall restrict attendance modification privileges to Instructor and Administrator roles only.

REQ-4: The system shall reject any authentication token that has exceeded its expiration time by returning an unauthorized access response.

REQ-5: The system shall log failed login attempts for security monitoring.

REQ-6: The system shall display a user-friendly error message for invalid login attempts.

Related User Stories:

US-1: As a student, I want to log into the system using a project-specific account, so my attendance can be tracked using campus authentication.

US-18: As a system administrator, I want to manage user roles and permissions so that students and instructors only have access to features relevant to their role

8.2 Face Enrollment and Biometric Registration

8.2.1 Description and Priority

This feature allows students to enroll their facial data for recognition during attendance sessions. The system captures multiple facial angles and generates encrypted beddings.

8.2.2 Stimulus/Response Sequences

1. Students select “Enroll Face.”
2. System Activates Camera.
3. Students follow guided prompts for head movement.
4. The system captures multiple images.
5. AI service generates an embedding.
6. Embedding is encrypted and stored.
7. A confirmation message is displayed.

8.2.3 Functional Requirements

REQ-1: The system shall capture at least three facial angles during enrollment.

REQ-2: The system shall convert captured images into numerical embeddings using the AI service.

REQ-3: The system shall encrypt facial embeddings before storage.

REQ-4: The system shall delete all captured facial images within 5 seconds after generating facial embeddings used for recognition.

REQ-5: The system shall allow administrators to re-enroll or delete biometric data.

REQ-6: The system shall prevent duplicate enrollment of the same student ID.

Related User Stories:

US-4: As a student, I want to enroll my facial data through the application so that the system can recognize me during the in-class attendance check-ins.

US-10: As a student, I want to scan a QR code displayed by the instructor, so that I can access the attendance or enrollment page without manually entering a URL.

US-11: As a student, I want to receive feedback when my face scan fails, so that I know whether to retry the scan or contact my instructor.

US-12: As a student, I want to update my facial data so that the system can continue to recognize me if my appearance changes.

US-13: As a student, I want to opt out of facial recognition, so my privacy concerns are respected.

8.3 Attendance Session Management

8.3.1 Description and Priority

This feature allows instructors to create, start, monitor, and end attendance sessions for specific courses.

8.3.2 Stimulus/Response Sequences

1. The instructor selects the course.
2. Instructor clicks “Start Session.”
3. The system creates a session ID and activates recognition.
4. Students check in.
5. The instructor monitors the real-time attendance list.
6. Instructor clicks “End Session.”
7. Session closes, and records are finalized.

8.3.3 Functional Requirements

REQ-1: The system shall generate a unique session identifier for each attendance session.

REQ-2: The system shall timestamp each check-in event.

REQ-3: The system shall prevent check-ins after session closure.

REQ-4: The system shall allow instructors to manually override attendance status.

REQ-5: The system shall reject any attempt to mark attendance more than once for the same student within a single session.

REQ-6: The system shall allow export of attendance records in CSV format.

REQ-7: The system shall allow instructors to configure the number of minutes after a scheduled class start time at which a student should be considered late.

Related User Stories:

US-5: As an instructor, I want to start and stop an attendance session for a specific class period, so attendance is only recorded during valid class times.

US-3: As an instructor, I want to manually mark or correct attendance so that errors or special cases can be handled.

US-8: As an instructor, I want to view attendance in real time during class so that I can immediately see who is present or absent.

US-6: As a student, I want to receive confirmation after a successful attendance scan, so I know my attendance was recorded correctly.

US-11: As a student, I want to receive feedback when my face scan fails, so that I know whether to retry the scan or contact my instructor.

8.4 Instructor Dashboard & Attendance Monitoring

8.4.1 Description and Priority

The Instructor Dashboard provides instructors with a centralized interface to monitor, manage, and review attendance sessions. Through this dashboard, instructors can view attendance records in real time, manually adjust attendance statuses, and identify students who have opted out of facial recognition.

Priority: High

8.4.2 Stimulus/Response Sequences

1. The instructor logs into the system
2. The system displays the Instructor Dashboard with available courses and session controls
3. Instructor starts an attendance session
4. The dashboard displays real-time updates as students are marked present.
5. Instructor manually edits a student's attendance status.
6. The system updates the record and logs the modification
7. Instructor selects a previous session
8. The system displayed the stored attendance records for that session.

8.4.3 Functional Requirements

REQ-1: The system shall provide instructors with a dashboard interface for managing attendance sessions.

REQ-2: The system shall display real-time attendance updates during active sessions.

REQ-3: The system shall allow instructors to manually modify attendance records.

REQ-4: The system shall display a list of students who have opted out of facial recognition.

REQ-5: The system shall allow instructors to search or filter attendance records by student name or session date.

REQ-6: The system shall provide instructors with dashboard control to view and update the late attendance threshold for a session.

Related User Stories:

US-2: As an instructor, I want to view attendance records in a dashboard so I can quickly see who is present and absent.

US-3: As an instructor, I want to manually mark or correct attendance so that errors or special cases can be handled.

US-8: As an instructor, I want to view attendance in real time during class so that I can immediately see who is present or absent.

US-14: As an instructor, I want to see which students have opted out of facial recognition, so that I can manage their attendance manually.

8.5 Attendance Reporting & History

8.5.1 Description and Priority

The Attendance Reporting & History feature enables long-term storage, retrieval, and export of attendance records. Instructors may export records for grading or documentation purposes, and students may view their own personal attendance history.

Priority: Medium-High

8.5.2 Stimulus/Response Sequences

1. The Instructor selects an attendance session and clicks export
2. The system generates a downloadable CSV file containing attendance records.
3. Students access their attendance history page.
4. The system displayed the student's attendance record by course and date.

8.5.3 Functional Requirements

REQ-1: The system shall store attendance records by course and session date.

REQ-2: The system shall allow instructors to export attendance records in CSV format.

REQ-3: The system shall allow students to view their personal attendance history.

REQ-4: The system shall ensure that students may only access their own attendance records.

Related User Stories:

US-15: As an instructor, I want to export attendance records as a CSV or PDF file, so I can maintain records or perform grade calculations outside the system.

US-16: As a student, I want to view my own attendance history in the system, so that I can track my attendance status throughout the semester.

8.6 Canvas Integration

8.6.1 Description and Priority

The Canvas Integration feature allows FaceRoll to retrieve course roster information from the Canvas Learning Management System. This ensures accurate student enrollment data for attendance sessions.

Priority: Medium

8.6.2 Stimulus/Response Sequences

1. Instructor links a course within FaceRoll.
2. The system retrieves the corresponding course roster from Canvas via API.
3. The instructor selects a Canvas-linked course when starting a session.
4. The system associates the attendance session with the selected course roster.

8.6.3 Functional Requirements

REQ-1: The system shall retrieve course roster data using the Canvas API.

REQ-2: The system shall associate attendance sessions with a Canvas course identifier

REQ-3: The system shall securely authenticate API requests to Canvas.

Related User Stories:

US-19: As an instructor, I want to link my Canvas course to FaceRoll so that the correct student roster is automatically imported for attendance sessions.

9.

9.1 Performance Requirements

PR-1: The system shall recognize and process a detected face within **3 seconds** under normal classroom conditions (standard lighting and ≤ 60 students).

PR-2: The system shall support attendance processing for a class size of up to **60 students per session** without system slowdown.

PR-3: The instructor dashboard shall load attendance data within **2 seconds** after a session begins.

PR-4: The system shall record attendance events in the database within **1 second** after successful recognition.

PR-5: The system shall maintain continuous operation for a minimum of **2 hours** (typical class duration) without performance degradation.

Rationale:

These performance requirements ensure attendance can be recorded in real time without interrupting classroom activities or causing delays for instructors.

9.2 Safety Requirements

SR-1: The system shall not permanently store raw video recordings from the classroom camera.

SR-2: The system shall allow instructors to manually correct attendance records to prevent academic harm caused by recognition errors.

SR-3: The system shall prevent accidental deletion of attendance records without confirmation.

SR-4: The system shall automatically save attendance data periodically to prevent data loss during unexpected shutdowns.

SR-5: The system shall comply with institutional privacy expectations and applicable educational data protection practices.

Rationale:

Incorrect attendance records could negatively affect student grades or participation tracking; safeguards ensure recoverability and correction.

9.3 Security Requirements

SEC-1: The system shall require authenticated login for instructor access to the dashboard.

SEC-2: All communication between client and server shall use HTTPS with TLS encryption.

SEC-3: Facial data shall be stored as encrypted facial embeddings, not raw images, whenever possible.

SEC-4: Only authorized instructors or administrators shall be permitted to modify attendance records.

SEC-5: The system shall maintain an audit log that records all attendance modifications, including timestamps and user IDs.

9.4 Software Quality Attributes

Reliability

- The system shall achieve at least 95% recognition accuracy under normal lighting conditions.
- The system shall continue operating even if a single recognition attempt fails.

Usability

- An instructor shall be able to learn basic system operation within 30 minutes without technical training.
- The interface shall use clear visual indicators for attendance status (Present, Late, Absent).

Maintainability

- Source code shall include documentation and comments sufficient for future capstone teams to maintain or extend the system.
- A modular architecture shall be used to separate the recognition, database, and interface components.

Testability

- Each functional requirement shall be verifiable through defined test cases.

9.5 Business Rules

BR-1: Only instructors are authorized to start or end attendance sessions.

BR-2: Only instructors or administrators may modify attendance records.

BR-3: Attendance must be associated with a specific class session and timestamp.

BR-4: Each student may only receive one attendance status per session.

BR-5: Manual attendance changes must be recorded in an audit log.

BR-6: Students do not have direct access to modify attendance data.

BR-7: Attendance records shall be retained for at least one academic semester unless removed by authorized personnel.

10. Other Requirements

10.1 Database Requirements

- The database will use Firebase Firestore
- All facial embeddings will be stored in encrypted format
- Attendance records will include:
 - Student ID
 - Course ID
 - Attendance status
 - Timestamp
- The system will retain data for at least a semester

10.2 Legal and Regulatory Requirements

- The system will comply with FERPA privacy standards

- Biometric data will not be shared
- Students must provide consent before biometric data is collected

10.3 Reusability and Maintainability

- Source code will be well documented
- Modular architecture will be used
- Components will be separate and distinct for future development

Appendix A: Glossary

Face Embedding – A numerical vector representation of a detected face used for comparison.

Session ID – A unique identifier assigned to an attendance session.

FERPA – Family Educational Rights and Privacy Act governing student data privacy.

OAuth 2.0 – Industry-standard authorization protocol used for Canvas API integration.

Recognition Confidence Score – Numerical value representing similarity between facial embeddings.

Role-Based Access Control – System method for restricting access based on assigned user role.

LMS- Learning Management System

API- Application Programming Interface

Appendix B: Analysis Models

The system architecture consists of:

Mobile Client (React Native App)

Backend Service (Node.js + Firebase)

AI Recognition Microservice (Python + OpenCV + FaceNet)

Firestore Database

Canvas API Integration

Primary Entity Relationships:

User → Enrollment → Course

Course → AttendanceSession → AttendanceRecord

User → FacialEmbedding

Appendix C: To Be Determined List

TBD-1: Final recognition confidence threshold value.

TBD-2: Maximum class size scalability limit beyond 60 students.

TBD-3: Institutional approval timeline for full Canvas production integration.

TBD-4: Final audit log retention duration.

TBD-5: Final decision on automatic attendance synchronization to Canvas gradebook